

Project Proposal:

Image Captioning using CNN-RNN Architecture

María Jaque Oficialdegui, Emma Rey Sánchez,
José Andrés Ridruejo Tuñón, Catalina Royo-Villanova Seguí

April 2024

1 Introduction

Image captioning represents a significant advancement at the intersection of computer vision and natural language processing (NLP), aiming to generate descriptive textual annotations for visual content. During the following pages we will discuss possible sources of data to train an image processing model as well as possible structures and evaluation metrics. Finally, we will explain our milestones for the future progress reports.

2 Possible Datasets

In the papers we have read and the online research we have carried out, we found the following datasets that will now be explained.

Firstly, the MS COCO (Microsoft Common Objects in Context) is a dataset containing 164K images with each five captions describing the scene among other annotations. The original split (in 2014) had 83K images in the training set, 41K for validation and another 41K in the testing set but in 2017 the train and validation sets were restructured so that the train dataset contains 118K images and the validation dataset contains 5K images. This last structure will be the one we use.

Then there are the Flickr8k and Flickr30k datasets, the second being an extension of the first. In these datasets there are also five captions for each image, but they have not a fixed split between train, validation, and test.

Finally, we have found some other datasets such as Google's Conceptual Captions dataset with more than three million images and SBU Captioned Photo Dataset with a million images. The problem with these datasets is that the captions may be subjective or not very accurate as in the Conceptual Captions dataset, the captions are obtained from the Alt-text HTML attribute of internet images and the SBU dataset consists of the image and its caption given by the owner when uploading it to Flickr. [5]

3 Model Approach: CNN-RNN

The CNN-RNN (Convolutional Neural Networks – Recurrent Neural Network) framework is highlighted as a prominent approach in the realm of image captioning, leveraging the strengths of both CNNs for image feature extraction and RNNs for sequential sentence generation. This combination allows for the effective translation of visual content into descriptive sentences, capturing not just the objects within an image but also their attributes and relationships. The union of both Neural Network is established via an encoder-decoder architecture.

The encoder-decoder structure is a foundational framework in the field of image captioning, particularly in the implementation of CNN-RNN. This structure effectively translates visual input into descriptive textual output, mirroring processes observed in machine translation from one language to another. The Convolutional Neural Network (CNN) will be the encoder, responsible for processing the input image and extracting a high-dimensional feature vector that encapsulates the visual information present in the image. This feature vector contains valuable, compressed representations of the image's content, including objects, attributes, and their spatial relationships. Some

papers report good results by using a ResNet50 [4] or a karpathy image captioning model [3]. Even though the models can be different in structure the main difference for the overall model is the way of including the encoded information into the decoder. For our specific project we aim to use a VGG19 net, that consists of a 19 layer convolutional network pretrained with the Imagenet dataset [2].

Following the encoding process, the decoder, in this case the Recurrent Neural Network (RNN) takes over to generate a coherent and contextually relevant sentence that describes the image. It is worth mentioning that some variants of RNN, like Long Short-Term Memory (LSTM) or Gated Recurrent Units (GRU) are good options of decoders that will be considered in this project. The decoder translates the complex feature vector provided by the encoder into a sequence of words, one at a time, until a complete sentence is formed.

In conclusion, this framework's success is underscored by its adaptability and efficiency in handling the dual challenges of understanding and translating visual content into descriptive language.

4 Model Evaluation

After reviewing the research papers, we have identified several types of caption evaluators. These evaluators aim to mimic human evaluation, which generally yields the best results, but is also the most expensive and time-consuming approach. Each evaluation method has its own advantages and disadvantages. Let's explore the different evaluators to ultimately choose the one that best suits our project:

- BLEU (Bi-Lingual Evaluation Understudy): This is the most common evaluator found in the papers we reviewed. It measures n-gram precision between generated captions and reference captions. While fast and efficient, BLEU struggles to account for synonyms and paraphrasing.
- ROUGE (Recall-Oriented Understudy for Gisting Evaluation): Similar to BLEU but focuses on recall. This approach seems to promote repetitive captions.
- METEOR (Metric for Evaluation of Translation with Ordering): METEOR combines BLEU with ROUGE and considers word order. However, it can be computationally expensive.
- CIDEr (Consensus-based Image Description Evaluation): CIDEr incorporates the BLEU score with word overlap and Rouge-L. It is sensitive to variations in the reference captions.

All these models have their weaknesses and are far from matching the results of human evaluation. New types of evaluation have emerged that improve accuracy but require more data processing [1]. This is why we decided on these approaches, CIDEr being the best of the four.

5 Steps and Milestones

To achieve a successful image captioning model that can be trained with the computational resources available, there are indispensable steps that must be taken. First, we will perform the data collection step, downloading the available data from the datasets mentioned above, as well as the pretrained model. Once the data is obtained, the next step will be the image and captions preprocessing. For this aim, all images will be transformed into RGB with a fixed size that is compatible with the model. Numbers and extra spaces will be removed from captions, and they will be turned to lower case sentences. Images without a caption will be dropped from the training, validation, and test sets to avoid ambiguity and start and end of sentence tokens will be added. In addition to this step, word-to-idx and idx-to-word dictionaries will be added to represent every distinct word in the vocabulary. To proceed we will define and fit the models for the images and the captions. Finetuning will be needed for the image model and an exhaustive analysis of the possible RNNs will be performed according to the evaluation metrics.

As for the temporal planning of the project, will achieve the following for each report:

- 12th April 2024: data collection, image and caption preprocessing and first model implementation.
- 19th April 2024: bug fix of the model and fine tuning of the parameters.

References

- [1] Yin Cui et al. “Learning to Evaluate Image Captioning”. In: (2018). URL: <https://arxiv.org/abs/1806.06422>.
- [2] G. Geetha et al. “Image Captioning Using Deep Convolutional Neural Networks (CNNs)”. In: *Journal of Physics: Conference Series* (2020). URL: <https://iopscience.iop.org/article/10.1088/1742-6596/1718/1/012065>.
- [3] Andrej Karpathy and Li Fei-Fei. “Deep visual-semantic alignments for generating image descriptions”. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2015, pp. 3128–3137.
- [4] Talakoti Mamatha et al. “Generating Image Captions Using Deep Learning Algorithms RNN and LSTM”. In: *Journal of Aeronautical Materials* (2023). URL: <http://www.hkclxb.cn/oa/pdftow.aspx?Sid=202317>.
- [5] Oriol Vinyals et al. “Show and tell: A neural image caption generator”. In: *CVPR*. 2015. URL: <https://arxiv.org/abs/1411.4555>.